Boundary-guided Black-box Fairness Testing

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***Abstract*—Although deep learning models have achieved outstanding performance in many applications, there are still concerns about their fairness. A series of fairness testing methods, which evaluate the fairness of deep learning models by generating discriminatory samples, have been proposed. However, these methods either neglect the naturalness of discriminatory samples or roughly select natural discriminatory samples, leading to a decrease in efficiency. In this paper, we introduce a boundary-guided black-box fairness testing method to effectively generate individual discriminatory samples with high efficiency and enhanced naturalness. Our boundary-guided method involves a global exploration phase, which explores multiple paths from the initial samples to the surrogate decision boundary of the target model, imitated from the semantic latent space of a generative adversarial network (GAN). Then, a local perturbation phase explores the nearby space around a given sample for identifying potential discriminatory samples. Extensive experiments on various datasets demonstrate that our approach outperforms state-of-the-art methods in terms of efficiency and effectiveness while maintaining high naturalness.**

***Index Terms*—Fairness Testing, Boundary-Guided Method, Individual Discriminatory Samples.**

I. Introduction

Deep learning has been widely applied in various decision-making scenarios in applications, such as criminal risk assessment [1], loan industry [2], and medical diagnosis [3]. Though deep neural networks (DNNs) exhibit exceptional performance in various decision-making tasks, their susceptibility to minor input data perturbations potentially triggers discrepancies in prediction outcomes upon alterations to specific data samples. For instance, the COMPAS crime risk assessment algorithm [4] developed by Northpointe disproportionately assigns lower crime risk scores to white individuals while assigning higher crime risk scores to black individuals at twice the rate of white individuals.

Since the scale and distribution of data in real datasets are insufficient to fully reflect the discriminatory behavior of the model, to evaluate the fairness (i.e. non-discrimination) [5] of DNNs that are adopted in various applications, numerous studies have been conducted to generate test samples that could potentially violate fairness requirements and reveal any existing biases in the model. Fairness evaluation is approached from two main perspectives: individual fairness [6], [7] and group fairness [8], [9]. Individual fairness measures whether individuals with only intrinsic or acquired differences are treated similarly in the decision-making process, while group fairness examines whether statistical parity exists across different groups. Evaluating the individual fairness of DNNs helps to reveal discriminatory behaviors that are overlooked by group fairness [7]. For individual fairness testing, most existing methods [7], [10]–[13] only focus on improving the efficiency and effectiveness of generating discriminatory samples, without considering whether the generated discriminatory samples resemble the data distribution of the original dataset or even violating constraints in the real world [14]. Xiao et al. [15] recognize this issue and propose a solution to improve the naturalness of discriminatory samples. Yet, it relies on approximating the real decision boundary of the target model by deriving a linear surrogate boundary. The discriminatory candidates are then coarsely generated on or near the surrogate boundary, which leads to a decrease in high-quality samples. To address this problem, we propose an effective, efficient, and naturalness-preserving approach to generate discriminatory samples for DNNs.

To achieve a good balance between efficiency, effectiveness, and naturalness of discriminatory sample generation, this work proposes a boundary-guided fairness testing approach. Our boundary-guided method identifies the direction where the boundary lies and explores along this direction to generate more discriminatory samples with high naturalness. We construct an auxiliary dataset that maps the latent space of a generative model and train it to obtain a linear hyperplane, serving as a surrogate decision boundary. Then, we explore multiple paths from the initial samples in the auxiliary dataset, guided by the distance to the hyperplane and the normal vector of the hyperplane. We can accelerate the exploration by optimizing a fitness score. To expand the search scope, we perform multiple rounds of exploration in the neighborhood of the samples. Therefore, we can quickly generate a large number of discriminatory samples with high naturalness.

To evaluate our boundary-guided approach, we conduct extensive experiments by generating discriminatory samples for DNNs on multiple real-world datasets. In comparison with two state-of-the-art individual fairness testing methods, including ExpGA [11] and LIMI [15], our boundary-guided approach generates $\times 2.77$ more individual discriminatory samples at $\times 2.62$ faster speed. Additionally, by ensuring efficiency improvements without sacrificing the naturalness (80.56% on average) of discriminatory samples, we are able to enhance the model's fairness in both individual fairness (23.84% on IF_o and 71% on IF_d) and group fairness (18.62% on AOD

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and 17.33% on SPD) by retraining. Our contributions are as follows:

- To the best of our knowledge, we are the first to guide individual discriminatory sample generation based on the spatial position of the decision boundary in black-box settings, which could help reveal the actual model fairness.
- We propose a boundary-guided fairness testing method that incorporates both a global exploration phase (referred to as GlobalExplore) and a local perturbation phase (referred to as LocalPerturb) to achieve a good balance between efficiency, effectiveness, and naturalness of individual discriminatory sample generation.
- Extensive experimental results demonstrate that our method outperforms existing SOTA methods largely in effectiveness ($\times 2.77$ samples) and efficiency ($\times 2.62$ speeds) on average, maintaining high naturalness (80.56% on average, with a gap of 0.13% compared to a SOTA method).

II. Preliminaries

In this section, we first provide a brief review of relevant background, including deep learning networks and generative adversarial nets. Then, we explain the problem definition.

A. Backgrounds and Notations

Deep neural networks. Given a dataset $\mathcal{D}$ with sample $x \in X$ and label $y \in Y$, a DNN with parameter θ is denoted as $f_\theta : X \to Y$. The layers in a DNN model are denoted as $NL = \{NL_i | i \in \{0, ..., N\}\}$ and the i-th layer has n_i neurons. A DNN computes the outcome of each neuron by applying an activation function ϕ (such as Sigmoid, Hyperbolic Tangent (tanh), or Rectified Linear Unit (ReLU) [16]) to the weighted sum of the outputs of all the neurons in its previous layer. The purpose of a DNN model for binary classification tasks is to learn a decision boundary, which is a surface in the input domain that can divide the inputs into two classes. The decision boundary is formulated as:

$$B = \{x : \theta(x) = 0\}, \tag{1}$$

where $\theta(x) < 0$ denotes x is assigned to the negative class, while $\theta(x) > 0$ indicates x is assigned to the positive class.

Generative adversarial networks. GANs are widely used to generate high-quality samples for data synthesis. The GAN framework was first proposed for image synthesis by Goodfellow et al. [17], which trains two adversarial neural networks, the generator G and the discriminator D. The generator G aims to mimic the given real training dataset distribution, which maps a random latent vector $\mathbf{z}$ to a synthetic sample $g(\mathbf{z})$, while the discriminator D strives to accurately distinguish between real and generated data. The generator G and the discriminator D form a dynamic game process. GANs undergo alternating optimization training, where both the generator G and discriminator D are improved. Ultimately, we aim to achieve a highly effective generator G that produces data indistinguishable from real data. The training process of GAN involves optimizing the following min-max objective:

$$\min_G \max_D E_{x\sim p_{data}}[logD(x)]+E_{z\sim p_z}[log(1-D(G(z)))], \tag{2}$$

where $x \sim p_{data}$ denotes the real data distribution, and $z \sim p_z$ denotes the distribution of latent space.

Besides images, GANs can be applied to generate various discrete data, such as tabular data. Xu et al. [18] first introduced TGAN, which can generate high-quality and fully synthetic tables while simultaneously generating discrete and continuous variables. Building upon this, Xu et al. [19] proposed CTGAN, which invents the mode-specific normalization to overcome the non-Gaussian and multi-modal distribution, and further designs to deal with the imbalanced discrete columns.

B. Problem Definition

We first define **individual discrimination** and **individual discriminatory sample**. We denote X as a data sample set and its set of attributes by $A = \{A_1, A_2, \cdots, A_n\}$. Assuming each attribute A_i has a valuation domain $\mathbb{I}_i$, the input domain is then $\mathbb{I} = \mathbb{I}_1 \times \mathbb{I}_2 \times \cdots \times \mathbb{I}_n$, which represent all possible combination of attribute valuations. **Individual discrimination** refers to unequal treatment, such as different model decisions between two similar individuals who only differ in their protected attributes. The protected attributes are predefined in specific domains such as gender, country, and age. We use $P \subset A$ to represent the set of protected attributes and $NP \subset A$ to represent the set of non-protected attributes. For each sample $x = \{x_1, x_2, \cdots, x_n\} \in X$ (x_i is the value of attribute A_i), we define x as an **individual discriminatory sample** for model f_θ when there exists another sample x' satisfying:

$$\begin{aligned} &\exists p \in P, s.t.\ x_p \neq x'_p; \\ &\exists q \in NP, s.t.\ x_q = x'_q; \\ &f_\theta(x) \neq f_\theta(x'). \end{aligned} \tag{3}$$

Moreover, we use (x, x') to represent an individual discriminatory sample pair for f_θ. Thus, our goal in this paper is to generate individual discriminatory samples for fairness testing, which might force the target model to make biased decisions and violate the requirement of individual fairness. Following common assumptions in black-box fairness testing [11], [12], [20], we assume that the tester is provided with data samples X, corresponding attributes space A, and protected attribute set NP, while the target model f_θ trained on dataset $\mathcal{D} = \{X, Y\}$ is a black-box model (without knowledge of internal information of models such as gradients).

Naturalness of generated samples. In addition to individual discrimination, we also consider the naturalness of generated discriminatory samples. In the context of tabular data, naturalness measures the extent to which a dataset reflects the underlying distribution of the data in the real world [21]. Specifically, naturalness is a metric that measures the degree to which a dataset matches the statistical characteristics and relationships, such as value distributions and attribute correlations,

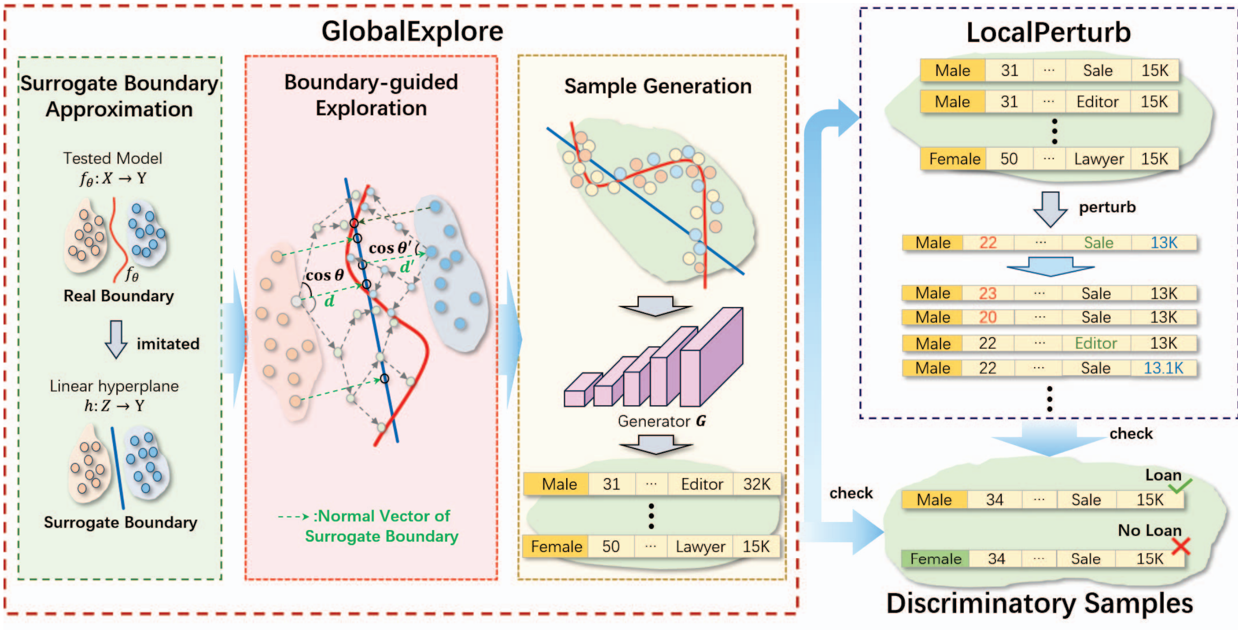


Fig. 1. Overview of the boundary-guided black-box fairness testing method

of the real-world data it represents [22]. Statistical metrics such as the Pearson's Correlation [23], Average Nearest Neighbor Distance [24], and Kolmogorov-Smirnov statistic [25] can be used to quantify the naturalness of the generated data. These metrics evaluate the distribution distance/similarity between the generated data and the original data, with the original data (real-world datasets) serving as the reference data distribution. Generated data with high naturalness closely resembles the statistical characteristics of the original dataset, indicating a more accurate representation of the data distribution [19]. Therefore, our goal is to generate highly natural individual discriminatory samples that accurately represent the data distribution, effectively revealing discriminatory behavior in real-world scenarios.

III. Methodology

In this paper, we propose a boundary-guided fairness testing framework to generate individual discriminatory samples for black-box fairness testing. To reach a good balance between efficiency, effectiveness, and naturalness of discriminatory sample generation, we design two generation phases: a global boundary-guided generation phase **GlobalExplore** and a local perturbation phase **LocalPerturb**. GlobalExplore explores multiple random paths toward the decision boundary of the target model to efficiently explore candidate discriminatory samples. LocalPerturb randomly perturbs the non-sensitive attributes of the candidate samples near the decision boundary to generate more discriminatory samples. Since the decision boundary of the target model often reflects the nature of the distribution of the original data and the test cases near the boundary are more likely to serve as discriminatory samples with slight perturbations and possess better naturalness, GlobalExplre allows us to quickly find samples distributed around the decision boundary, and LocalPerturb helps generate a large number of samples similar to those near the decision boundary. The overall framework is illustrated in Fig. 1.

A. Global Boundary-guided Exploration

Existing studies [26], [27] indicate that the decision boundary of the trained target model is often close to the majority of natural data samples and highly reflects the properties of the original data distribution. Meanwhile, applying minor perturbations to samples near the decision boundary with protected attributes is more likely to result in changes to the decision outcomes. Hence, generating test cases under the guidance of the decision boundary of the target model can evidently be more efficient and natural (closer to the original data distribution). However, in the black-box fairness testing scenario, we can not directly access internal model information or deduce the decision boundary in the original input domain. Thus, we attempt to approximate the decision boundary of the target model by deriving a surrogate boundary in the latent space of the GAN based on a constructed synthetic latent dataset. This allows us to obtain the spatial relationship between the decision boundary and the initial samples, enabling us to search for high-quality test cases near the decision boundary.

The logic of the global boundary-guided exploration phase GlobalExplore is shown in Algorithm 1. It takes the target model f_θ, the generator G of a GAN model that generates synthetic samples, the number of iterations for global exploration n_{global}, the number of exploration paths $p_{explore}$ and a threshold c_{thr} for the cosine value between unit vector and normal vector as inputs. The output is a set G_{discr} containing discriminatory samples.

Algorithm 1: Global Boundary-guided Exploration Phase

Input : model f_θ, generator G, n_{global}, $p_{explore}$, threshold $c_{thr} = 0.7$
Output: discriminatory sample set G_{discr}

1 $G_{discr} = \emptyset$
2 $h, \mathfrak{D}_{au} = GetSurrogateBoundary(f_\theta, G)$ //$h : \mathbf{w}^\top \mathbf{z} + b = 0$
3 **foreach** *vector set* V_{init} *in* $\mathfrak{D}_{au}$ **do**
4 $\quad V_{cur} = V_{init}$
5 $\quad X_{cur} \leftarrow G(V_{cur})$
6 $\quad \Delta = getSelectedProb(fit(X_{cur}))$
7 $\quad$ **for** i *in* $1 : n_{global}$ **do**
8 $\quad\quad V_{new} = \emptyset$
9 $\quad\quad$ **while** $|\, V_{new} \,| \leq (|\, V_{cur} \,| * p_{explore})$ **do**
10 $\quad\quad\quad \mathbf{uv} = InitUnitVector()$
11 $\quad\quad\quad$ **if** $CaculateCosin(\boldsymbol{uv}, \mathbf{w_u}) \geq c_{thr}$ **then**
12 $\quad\quad\quad\quad d_{step} = GetStepSize(fit(X_{cur}))$
13 $\quad\quad\quad\quad V_{new} \leftarrow V_{new} \cup (V_{cur} + \mathbf{uv} * d_{step} * (\mathbf{w_u}^\top \cdot V_{init} + b_u))$
14 $\quad\quad\quad$ **end**
15 $\quad\quad$ **end**
16 $\quad\quad V_{new} \leftarrow V_{new} \cup V_{cur}$
17 $\quad\quad X_{new} \leftarrow G(V_{new})$
18 $\quad\quad \Delta = getSelectedProb(fit(X_{new}))$
19 $\quad\quad V_{expl} = \emptyset$
20 $\quad\quad X_{expl} = \emptyset$
21 $\quad\quad$ **while** $|\, V_{expl} \,| \leq |\, V_{cur} \,|$ **do**
22 $\quad\quad\quad s, x = Selectsample(V_{new}, X_{new}, \Delta)$
23 $\quad\quad\quad V_{expl} \leftarrow V_{expl} \cup s$
24 $\quad\quad\quad X_{expl} \leftarrow X_{expl} \cup x$
25 $\quad\quad$ **end**
26 $\quad\quad V_{cur} = V_{expl}$
27 $\quad\quad$ **for** *sample* x *in* X_{expl} **do**
28 $\quad\quad\quad$ **if** x *is an individual discriminatory sample* **then**
29 $\quad\quad\quad\quad G_{discr} \leftarrow G_{discr} \cup x$
30 $\quad\quad\quad$ **end**
31 $\quad\quad$ **end**
32 $\quad$ **end**
33 $\quad G_{discr} \leftarrow G_{discr} \cup LocalPerturb(X_{expl})$
34 **end**
35 **return** G_{discr}

GAN has been widely used to synthesize large amounts of realistic data for software testing [28], [29]. Therefore, we utilize GANs to help us generate a synthetic latent dataset. This synthetic latent dataset maps the latent space to the model decision space, enabling us to better obtain the surrogate decision boundary. As studies [30], [31] have shown that the latent space of GANs is approximately linearly separable for any binary semantic attribute (e.g., loan or no loan), we use a linear hyperplane as the rough surrogate decision boundary. The normal vector of this hyperplane is perpendicular to it, clearly indicating the position of the decision boundary in the latent space, thereby effectively guiding the generation of test samples near the decision boundary. Specifically, we first use the function $GetSurrogateBoundary$ (line 2) to approximate a linear hyperplane h that serves as the surrogate decision boundary in the latent space of GAN [15].

The hyperplane h is formulated as

$$h : \mathbf{w}^\top \mathbf{z} + b = 0, \tag{4}$$

where $\mathbf{w}$ denotes the normal vector of the hyperplane, b denotes the intercept term. We randomly sample a set of latent vectors $\mathbf{z}$ in the latent space to construct an auxiliary dataset $\mathfrak{D}_{au} = \{\mathbf{z} | \mathbf{z} \in Z\}$. A linear classification model is then trained on the samples corresponding to vectors in $\mathfrak{D}_{au}$ to approximate the parameters $\mathbf{w}$ and b.

With the obtained surrogate boundary h, we can easily calculate the distance d from any random latent vector $\mathbf{z}$ to h, as show below:

$$d = \frac{\mathbf{w}^\top \mathbf{z} + b}{\sqrt{\mathbf{w}^\top \mathbf{w}}}. \tag{5}$$

The latent vector $\mathbf{z}$ is assigned to an unfavorable sample (e.g., not loan) of model f_θ when it satisfies $d < 0$, otherwise, a favorable sample (e.g., loan) if it satisfies $d > 0$. Therefore, by moving in the same direction as the vector $\mathbf{w}\frac{d}{|d|}$, we can gradually approach the real boundary in the latent space of the GAN. Furthermore, since we do not directly generate samples on the surrogate decision boundary h, we do not need to be concerned about the fitness [15] between h and the real boundary.

Existing studies [13], [32], [33] have revealed the fact that data samples near the decision boundary are more likely to mislead the model's predictions with only minor perturbations. Thus, after obtaining the surrogate decision boundary, we start exploring in the latent space from random initial samples, moving toward the decision boundary of the target model, to find potential natural and discriminatory latent vectors. Concretely, in the main loop (lines 3 to 34), we randomly sample a batch of vector set V_{init} from the auxiliary dataset $\mathfrak{D}_{au}$ and explore paths from the samples in V_{init} towards the real boundary with the guidance of the normal vector $\mathbf{w}$. To generate the discriminatory samples with high effectiveness, we design a fitness score $fit(x)$, and a higher fitness score means the sample x is more likely to be a discriminatory sample. It is formulated as:

$$fit(x) = 1 - |Prob(x, l) - Prob(x, l')|, \tag{6}$$

where $Prob(x, l)$ denotes the probability of x being predicted as label l. We map the latent vectors in V_{cur} to synthetic samples X_{cur} and calculate the probability distribution Δ of each sample in X_{cur} to be selected in the new sample set.

The global exploration process is repeated for n_{global} times (lines 7 to 32). In each exploration step, the function

$GetStepSize$ provides an exploration step size d_{step} according to the average fitness score of X_{cur} (line 12). We randomly generate a unit vector **uv** as the exploration direction (line 10). If the cosine of the angle between **uv** and the normal vector **w** exceeds the threshold c_{thr}, we calculate a displacement vector according to the exploration step size d_{step} and exploration direction **uv** to get a new set of vectors V_{new} (lines 11 to 14). We then update the selection probability distribution Δ using the fitness score (lines 16 to 18) and select a sample set X_{expl} for the next round of exploration according to Δ (lines 21 to 25). We check whether the samples in X_{expl} are discriminatory samples by modifying the selected protected attributes of the samples, and the discriminatory samples are added to the set G_{discr} (lines 27 to 31). Finally, after n_{global} times of global exploration, we leverage the $LocalPerturb$ process to obtain more discriminatory samples from the generated high-quality sample set that is closer to the boundary (line 33).

B. Local Perturbation

Algorithm 2 describes the local perturbation phase LocalPerturb. Its goal is to generate numerous potential discriminatory samples distributed around a sample x. Its inputs include a sample set X, the number of local perturbations n_{local}, the perturb column scale c_{pert}, and the perturb value scale v_{pert}. The output is a local discriminatory sample set L_{discr}.

For every sample x in the sample set X, the local perturbation processing iterates n_{local} times (lines 3 to 9). In each iteration, the $GetSelectCols$ function randomly selects a subset C_{slt} of non-protected attribute columns based on the perturbation column ratio c_{pert} (line 4) and the function $PerturbValue$ randomly perturb the column values of x to obtain a perturbed sample x_{pert} based on the perturbation value ratio v_{pert} for each column in C_{slt} (line 5). We determine if x_{pert} is a discriminatory sample by modifying the values of the selected protected attribute. Once x_{pert} is identified as a discriminatory sample, it is added to the discriminatory sample set L_{discr} (lines 6 to 8).

Algorithm 2: Local Perturbation Phase

Input : sample set X, n_{local}, perturbation column ratio c_{pert}, perturbation value ratio v_{pert}
Output: discriminatory sample set L_{discr}

1 $L_{discr} = \emptyset$
2 **for** *sample* x *in* X **do**
3   **for** i *in* $1 : n_{local}$ **do**
4         $C_{slt} = GetSelectCols(c_{pert})$
5         $x_{pert} = PerturbValue(x, C_{slt}, v_{pert})$
6   **if** x_{pert} *is an individual discriminatory sample* **then**
7             $L_{discr} \leftarrow L_{discr} \cup x_{pert}$
8   **end**
9   **end**
10 **end**
11 **return** L_{discr}

IV. Evaluation

In this section, we evaluate the performance in terms of effectiveness, efficiency, the naturalness of generated samples, and the utility of fairness improvement.

A. Experiment Setup

Datasets. We evaluate our boundary-guided fairness testing method on four public tabular datasets, which are widely employed in the software fairness testing literature [11], [15]. Their descriptions are shown below:

- **Census Income Dataset [34]** identifies whether an adult's income exceeds \$50,000. It consists of 32,561 training samples and 16,281 testing samples. It has 13 features, including three protected attributes: gender, age, and race.
- **German Credit Dataset [35]** is used to assess the credit level of applicants (i.e., good or bad). It contains 600 samples described by 20 features. The protected attributes are gender and age.
- **Bank Marketing Dataset [36]** is derived from a direct marketing campaign conducted by a Portuguese banking institution. It is used to predict whether a customer will subscribe to a term deposit. It consists of 45,211 samples with 16 features, and the protected attribute is age.
- **Medical Expenditure Panel Survey Dataset** [1] aims to predict medical needs based on information provided. It includes 15,675 samples and has 40 attributes, and its protected attribute is gender.

For convenience, we will use the abbreviations Census, Credit, Bank, and Meps to refer to these four tabular datasets.

Baselines. We select two state-of-the-art approaches, i.e., ExpGA [11] and LIMI [15], as the baseline approaches. These methods are implemented using the code provided by the authors, and we follow the optimal hyperparameter settings in the original papers.

Evaluation Metrics. We use DPS, SUR, and ATN [15] to measure the **efficiency**, **effectiveness**, and **naturalness** of discriminatory sample generation, respectively. A discriminatory sample generation method is better if it achieves higher values of DPS, SUR, and ATN. We use IF_d, IF_o, SPD (Statistical Parity Difference) [37] and AOD (Average Odds Difference) [8] to measure **fairness**. We assume that a discriminatory sample detection method presents a better performance if it achieves smaller values of IF_o, IF_d, SPD, and AOD.

DPS denotes the speed of individual discriminatory sample generation,

$$DPS = \frac{DIN}{Time}, \tag{7}$$

where Time and DIN denote the consumed time and the number of detected discriminatory instances, respectively. SUR denotes the success rate for discriminatory samples,

$$SUR = \frac{DIN}{TN}, \tag{8}$$

[1]The Meps dataset: https://meps.ahrq.gov/mepsweb/data_stats/download_data_files_detail.jsp?cboPufNumber=HC-192

TABLE I
EFFECTIVENESS AND EFFICIENCY OF DIFFERENT GENERATION METHODS ON DIFFERENT DATASETS AGAINST DNN MODELS.

Dataset	Pro. Attr.	ExpGA				LIMI				Ours			
		DIN↑	TN	DPS↑	SUR↑	DIN↑	TN	DPS↑	SUR↑	DIN↑	TN	DPS↑	SUR↑
Census	gender	145,929	381,510	354.20	38.25%	79,228	856,263	187.30	9.25%	**279,223**	594,085	**612.33**	**47%**
	race	164,810	276,751	322.13	50.59%	147,626	751,272	206.76	19.65%	**342,368**	575,722	**526.72**	**59.47%**
	age	111,931	166,800	98.15	67.10%	189,866	638,755	199.02	29.72%	**395,651**	497,925	**867.66**	**79.46%**
Credit	gender	86,850	202,140	281.61	42.97%	164,569	1,000,000	358.91	16.46%	**359,797**	713,716	**634.56**	**50.41%**
	age	50,237	74,385	92.78	67.54%	459,750	1,000,000	373.26	45.98%	**456,977**	645,596	**490.35**	**70.78%**
Bank	age	27,385	50,461	69.68	54.27%	296,016	1,000,000	212.22	29.60%	**705,194**	1,068,358	**486.34**	**66.01%**
Meps	gender	258,226	533,350	193.06	**48.42%**	111,194	1,000,000	202.91	11.12%	**420,746**	1,182,628	**471.16**	35.58%

where TN denotes the total number of generated samples.

ATN denotes the distribution similarity between generated samples and the original dataset, which considers both the distribution of each column and the correlation between pairs of columns.

$$\begin{aligned} ATN(DIN, X) = & \frac{1}{2}[mean(KS(DIN, X_p) \\ & \cup TV(DIN, X_p)) \\ & + mean(PC(DIN_p, X_{p,q}) \\ & \cup CS(DIN_{p,q}, X_{p,q}))], \end{aligned} \tag{9}$$

where $mean(\cdot)$ calculates the average value of set, X_p denotes the p-th column of X, KS is an inverted version of the Kolmogorov-Smirnov Test. PC, CS, and TV denote the Pearson Coefficient, the Contingency Similarity, and the Total Variation Distance, respectively.

We follow previous work [12] to measure individual fairness, IF_d denotes the ratio of discriminatory samples in a randomly sampled discriminatory sample set, and IF_o denotes the proportion of individual discriminatory samples existing in the original dataset.

$$\begin{aligned} IF_d &= \frac{DIN}{|random\, sampling\, set|}, \\ IF_o &= \frac{DIN}{|original\, dataset|}. \end{aligned} \tag{10}$$

The SPD requires that a decision should be independent of the protected attributes [37], which measures the difference in positive classification between different demographic groups:

$$\begin{aligned} SPD = & |P(\hat{y} = 1 | A = unprivileged) \\ & - P(\hat{y} = 1 | A = priviliged)| \end{aligned} \tag{11}$$

where $\hat{y}$ denotes the model prediction and A is the group of the protected attribute.

The AOD represents the average of the differences in True Positive Rate (TPR) and False Positive Rate (FPR) between privileged and unprivileged groups [8]:

$$\begin{aligned} AOD = & \frac{1}{2}[|FPR_{A=unprivileged} - FPR_{A=privileged}| \\ & + |TPR_{A=unprivileged} - TPR_{A=privileged}|]. \end{aligned} \tag{12}$$

Implementation details. Our approach involves four important hyperparameters, including $p_{explore}$ and c_{thr} for exploration direction selection, n_{global}, and n_{local} for discriminatory sample generation.

We set c_{thr} as 0.7, c_{pert} as 0.2, v_{pert} as 0.1, and $p_{explore}$ as 3 to ensure the scope and efficiency of exploration. Without specification, we set n_{global} as 2 and n_{local} as 50. In the function $getSurrogateBoundary$, we follow LIMI [15] to obtain h and D_{au}. Our experiments are run on a personal computer with 24GB RAM, AMD Ryzen 7 5800H 3.20GHz CPU, and NVIDIA GTX3060 GPU in Windows 10.

B. Effectiveness and Efficiency

For the baselines, we follow their original settings [11], [15]. As for our approach, we select partial data (contains 30,000 latent vectors) from $\mathfrak{D}_{au}$ as initial data. All times are calculated only for inference time, i.e., the time taken for sample generation, excluding model training time.

The results on DNN are shown in Table I. In order to effectively compare the efficiency and effectiveness of our methods against the baseline, we inserted a timer into each method to count the discriminatory samples generated every five minutes and controlled the running time of all methods at the same level. From the results, we could make several observations as follows:

- *As for generation effectiveness,* our boundary-guided method generates more individual discriminatory samples than other baselines and outperforms them largely (+259,021 on average). As shown in Table I, the average DIN on four datasets of our approach is 422,851, which is **2.0 times** and **3.5 times** better than LIMI and ExpGA, respectively. Moreover, the average SUR on four datasets of our approach is 58.39%, which is **1.1 times** and **2.5 times** better than ExpGA and LIMI, respectively. In most datasets, our method is the best, except for a few cases. For example, on the Meps dataset, the SUR by our method is 35.58% while SUR by ExpGA is 48.42%. The above results demonstrate the outstanding performance of our approach on the test sample generation effectiveness, which is attributed to our boundary-guidance strategy and random perturbation strategy.
- *As for generation efficiency,* our boundary-guided method has a faster generation speed and outperforms other baselines largely (2.6 times on average). For instance, as shown in Table I, our method generates almost **584.16 discriminatory samples per second** on average while the value of ExpGA and LIMI is 201.66 and 248.63, respectively. This indicates that our fairness testing method can provide rapid feedback to testers, which will be

more suitable for software fairness testing tasks with time constraints. We conjecture that the primary factors of efficiency improvements are the random step size generation and the verification of exploration direction in GlobalExplore. These strategies serve to mitigate additional exploration costs and circumvent excessive exploration iterations.

In summary, our method presents the best efficiency and effectiveness. On average, our method generates $\times$**2.77** more individual discriminatory samples at $\times$**2.62** faster speed.

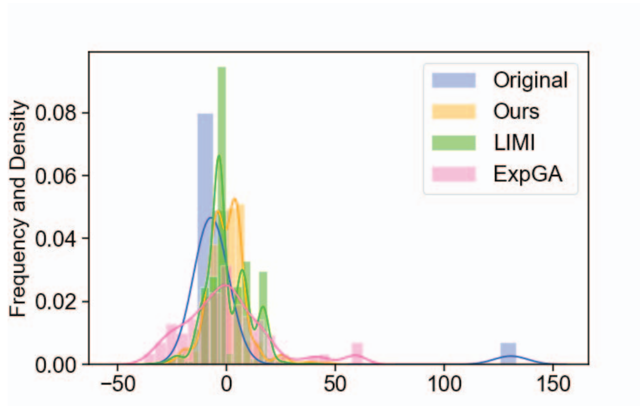
C. Naturalness of Discriminatory Samples

Naturalness measures the similarity in distribution between generated individual discriminatory samples and the original samples. We randomly select the same number of samples as the original dataset from an individual discriminatory sample set G_{discr} and then measure ATN for each set. From the results in Table II, we observe that the ATN value of our approach is much higher than that of ExpGA, and remains at the same high level as LIMI. Specifically, the ATN values of our method in all settings are around 80%, indicating that the generated samples by our method have high naturalness. In other words, the distribution of generated samples is highly similar to the original data distribution. This observation explains the outstanding performance of our method as it generates test samples close to the real decision boundary. In more detail, the average naturalness of ExpGA and LIMI on all datasets are 56.69% and 80.69%, and our method is 80.56%. We believe that the use of crossover and mutation operators in ExpGA contributes to its low naturalness. Both our method and LIMI exhibit a higher level of naturalness, as we both utilize the semantic latent space implicitly constrained by GAN. The random strategy used in LocalPerturb results in slightly lower naturalness of our method compared to LIMI.

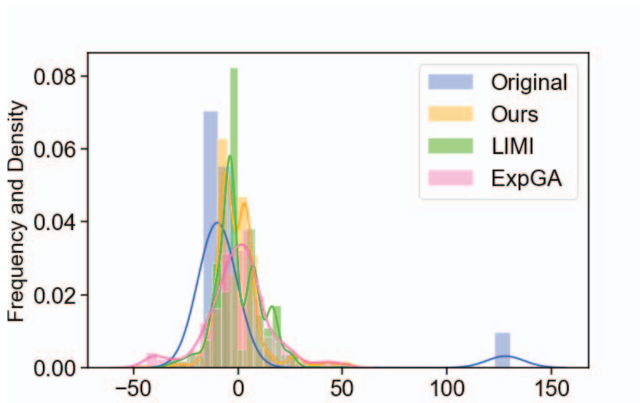
TABLE II
NATURALNESS OF DISCRIMINATORY SAMPLES GENERATED FOR DNN MODELS

Methods	Census			Credit		Bank	Meps
	gender	race	age	gender	age	age	gender
ExpGA	50.18%	47.73%	52.21%	63.35%	62.94%	57.08%	63.32%
LIMI	**81.54%**	81.24%	**81.34%**	**80.21%**	79.90%	79.51%	**81.78%**
Ours	81.41%	**81.77%**	80.16%	77.69%	**81.10%**	**82.08%**	79.69%

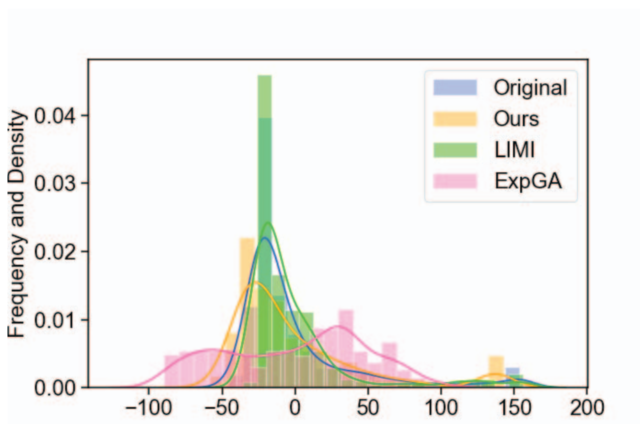
To better compare the naturalness, we visualize the feature similarity between the generated and original samples for ExpGA, LIMI, and our method. Specifically, we randomly select 1,000 discriminatory samples (600 for Credit) and employ PCA [38] to transform their feature into one dimension, we then visualize the feature distribution of these samples. As shown in Fig. 2, we can observe that the distribution of samples generated by our method (shown in orange) and LIMI (shown in green) are closer to the distribution of original data compared to ExpGA (shown in pink), which indicates that our method preserves the naturalness of discriminatory samples.



(a) Census-gender



(b) Census-age



(c) Bank-age

Fig. 2. Visualization of the data distribution of generated samples.

D. Fairness Improvement

To further demonstrate the validity of the discriminatory samples we generate, we follow [13] and retrain the DNN model using the generated discriminatory samples to improve model fairness. Since the quantity of the generated samples is much larger than the original dataset, we randomly selected

TABLE III

FAIRNESS IMPROVEMENT EXPERIMENTS VIA RETRAINING DNN MODELS.

Dataset	Pro. Attr.	Original					ExpGA					LIMI					Ours				
		ACC ↑	IF_o ↓	IF_d ↓	SPD ↓	AOD ↓	ACC ↑	IF_o ↓	IF_d ↓	SPD ↓	AOD ↓	ACC ↑	IF_o ↓	IF_d ↓	SPD ↓	AOD ↓	ACC ↑	IF_o ↓	IF_d ↓	SPD ↓	AOD ↓
Census	gender	88.11%	**0.08**	1.0	0.16	**0.03**	87.34%	0.1	0.27	0.17	0.04	87.37%	**0.08**	0.22	**0.14**	0.05	87.74%	**0.08**	**0.18**	0.15	**0.03**
	race	88.11%	0.16	1.0	0.11	0.04	87.76%	0.14	0.37	0.12	0.03	87.61%	**0.13**	**0.28**	**0.10**	**0.02**	88.10%	**0.13**	**0.28**	**0.10**	0.03
	age	88.11%	0.29	1.0	0.22	0.11	88.00%	0.29	0.69	0.25	0.15	87.50%	**0.21**	0.5	0.22	0.12	87.66%	**0.21**	**0.4**	**0.21**	**0.09**
Credit	gender	100%	0.14	1.0	0.07	**0.00**	95.50%	0.15	0.31	**0.06**	0.01	100%	0.13	**0.15**	0.07	**0.00**	100%	**0.08**	0.20	**0.06**	**0.00**
	age	100%	0.37	1.0	**0.00**	**0.00**	100%	0.35	0.54	**0.00**	**0.00**	100%	**0.34**	**0.5**	**0.00**	**0.00**	100%	**0.34**	**0.5**	**0.00**	**0.00**
Bank	age	93.85%	0.24	1.0	0.35	0.13	93.09%	0.22	0.57	0.29	0.11	92.82%	0.12	0.37	0.28	0.11	93.10%	**0.11**	**0.28**	**0.22**	**0.06**
Meps	gender	95.02%	0.13	1.0	0.14	0.03	92.41%	**0.11**	0.26	**0.07**	0.03	93.51%	**0.11**	0.22	0.09	0.03	94.65%	**0.11**	**0.19**	**0.07**	**0.02**

20% of the original dataset for retraining. We merge the original dataset with the selected discriminatory sample set to retrain the DNN model and evaluate the fairness of the retrained model with the fairness metrics (i.e. IF_o, IF_d, AOD, and SPD).

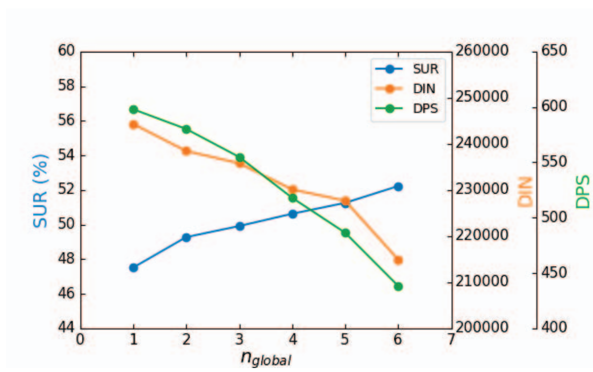
The results for individual and group fairness are shown in Table III, where the column "Original" represents the model trained on the original dataset, and the columns named by the corresponding methods represent models retrained with the discriminatory samples generated by each method. From the results, we have the following observations:

- For accuracy, the retrained models have a slight decrease (less than 1% on average except ExpGA on Credit with protected attribute gender), which indicates that the retrained models still have strong performance in real-world datasets.
- As for individual fairness, our method outperforms the baseline approaches significantly in terms of IF_o and IF_d, indicating that individual discriminatory samples generated by our method are more representative and have a greater impact on model decision-making. Specifically, our method achieves individual fairness improvement of 23.84% in terms of IF_o, and 71% in terms of IF_d, which achieves 17.60 times and 1.25 times than ExpGA, and 1.31 times and 1.04 times than LIMI on under these two individual fairness. Moreover, on two datasets, Credit and Census, models retrained by ExpGA even obtain worse individual fairness evaluation results in terms of IF_o.
- As for group fairness, we observe that the retained models achieve fairness improvements on almost all datasets. However, the improvements are not as significant as expected (+18.62% in terms of AOD, and +17.33% in terms of SPD). For example, on the dataset Census, our method achieves 6.63% and 14.39% improvement of SPD and AOC, respectively. By contrast, retraining by ExpGA and LIMI even make the models more unfair between different groups on Census and Credit. We speculate that the main reason is that samples generated by ExpGA deviate from the original data distribution. Although the distribution of samples generated by LIMI is closer to the original dataset, the disparity between the surrogate boundary and the real boundary leads to the difference between the generated samples and the original data. Similar to LIMI, we also observe overfitting on the small dataset Credit, and the model even behaves with no discrimination. Therefore, directly retraining models with new discriminatory samples would not increase fairness.

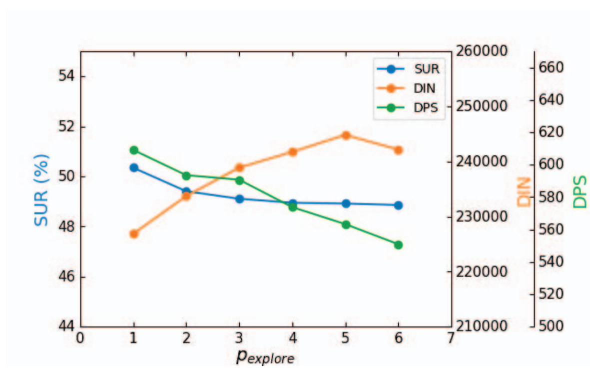
The discriminatory samples generated by our method are useful for fairness improvement. On average, we achieve individual fairness improvements of **23.84%** on IF_o and **71%** on IF_d, and group fairness improvements of **18.62%** on AOD and **17.33%** on SPD.

E. Hyperparameter Analysis

In addition to the main experiments conducted earlier, we further investigate and analyze hyperparameters to better understand our proposed method. In the global exploration phase, both the global exploration count n_{global} and exploration path count $p_{explore}$ are crucial. Thus, we investigate their impact on performance metrics. Specifically, we fix one and set the other as 1, 2, 3, 4, 5, and 6, respectively. The results are shown in Fig.3.



(a) n_{global}



(b) $p_{explore}$

Fig. 3. The experiment results of hyperparameter analysis.

- For n_{global}, we observe an increase in SUR with the rise of it whereas DIN and DPS decrease as it increases, as shown in Fig 3(a). A larger n_{global} results in the sample set being closer to the boundary, making it more likely to find discriminatory samples. However, it also constrains the exploration range, increasing the time cost of exploration. Considering the magnitude of changes in all metrics, we set $n_{global} = 2$ in our main experiments.
- For $p_{explore}$, we observe that DIN initially increases with its growth, then decreases when it reaches 6, while SUR and DPS decrease with the increase in it, as shown in Fig 3(b). A larger $p_{explore}$ widens the exploration scope but comes with increased exploration costs. Considering the magnitude of changes in all metrics, we set $p_{global} = 3$ in our main experiments.

V. Threats to Validity

Multi-class classification Model. We only verify the effectiveness of our method on the binary classification models. However, for multi-class classification models, the decision boundary is more complex and can even be closed, making it more challenging to determine the optimal exploration direction.

Protected Attribute. We only consider the single protected attribute in our experiments. Although multiply protected attributes fairness testing does not decrease the performance of our method, it introduces additional challenges to the assessment and analysis of discrimination.

VI. Related Work

In this section, we review the development in related fields.

A. *Fairness Testing.*

Recently, software fairness testing on machine learning models has received much attention. Fairness testing methods can be roughly divided into white-box methods and black-box methods. White-box methods [10], [13] have access to the internal structure and parameters of the model. In contrast, black-box methods [7], [11], [12], [15], [20] assess the model solely based on its inputs and outputs. In this paper, we focus on black-box fairness testing which is closer to real-world scenarios.

For **white-box fairness testing**, Zhang et al. [13] proposed ADF, which combines a global phase and a local phase to systematically search the input space for discriminatory samples with the guidance of gradient. Following ADF, Zhang et al. [10] proposed EIDIG, achieving better efficiency by introducing momentum into the global phase and using a limited budget for gradient calculation in the local generation phase.

For **black-box fairness testing**, Galhotra et al. [7] proposed THEMIS, which first defines software fairness testing, then introduces fairness scores as measurement metrics and lastly designs a causality-based algorithm randomly generating test inputs to evaluate the model fairness, i.e., the frequency of discriminatory samples' occurrence of software. However, THEMIS is inefficient in general since it relies on random sampling without guidance on the generation. To address this limitation, Udeshi et al. [12] introduced AEQUITAS, a systematic generating algorithm. It first explores the input domain randomly to discover discriminatory samples in the global search phase. During the local generation, AEQUITAS searches the neighbors of discriminatory samples identified in the global phase, by perturbing them. For the local generation, AEQUITAS designs three different strategies, i.e., random, semi-directed, and fully-directed, to update the probability which is used to guide the selection of attributes to perturb. Subsequently, Aggarwal et al. [20] presented SG, which integrates symbolic execution and local model explanation techniques to craft discriminatory samples. SG relies on the local explanation of a given input which constructs a decision tree utilizing the samples generated randomly by the Local Interpretable Model-agnostic Explanation (LIME). Fan et al. [11] proposed ExpGA, an explanation-guided method through the GA for software fairness testing. ExpGA uses interpretable methods to collect high-quality test seeds globally and employs the genetic algorithm to generate discriminatory offspring rapidly. Xiao et al. [15] proposed LIMI, which derives a surrogate linear boundary, then manipulates random latent vectors to the surrogate boundary with a one-step movement, and further conducts vector calculation to probe two potential discriminatory candidates.

B. *Boundary-Guided Adversarial Attacks*

Boundary guidance has been widely applied in adversarial attacks. Brendel et al. [39] introduce the Boundary Attack, a decision-based attack that starts from a large adversarial perturbation and then seeks to reduce the perturbation while staying adversarial. Chen et al. [40] develop HopSkipJumpAttack, a family of algorithms based on a novel estimate of the gradient direction using binary information at the decision boundary. Qin and Yue [41] design an AdvFuzzer to explore multiple paths between a source image and a guidance image, and design a LocalFuzzer to explore the nearby space around a given input for identifying potential adversarial examples.

VII. Conclusion

This paper proposes a boundary-guided fairness testing method to generate individual discriminatory samples. Our method combines a global boundary-guided exploration phase and a local perturbation phase to systematically explore the individual discriminatory samples close to the decision boundary using the direction of the surrogate linear boundary normal vector as guidance. In this way, we can rapidly approach the real decision boundary to generate numerous individual discriminatory samples while maintaining high naturalness. Extensive experiments demonstrate that our boundary-guided method generates natural discriminatory samples with a higher speed than state-of-the-art fairness testing methods in multiple real-world benchmarks. Augmented with the discriminatory samples generated by our method, the fairness of the tested models has substantially improved after retraining.

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